

High Availability

Availability %	Downtime per year ^[note 1]	Downtime per quarter	Downtime per month	Downtime per week	Downtime per day (24 hours)
90% ("one nine")	36.53 days	9.13 days	73.05 hours	16.80 hours	2.40 hours
95% ("one nine five")	18.26 days	4.56 days	36.53 hours	8.40 hours	1.20 hours
97% ("one nine seven")	10.96 days	2.74 days	21.92 hours	5.04 hours	43.20 minutes
98% ("one nine eight")	7.31 days	43.86 hours	14.61 hours	3.36 hours	28.80 minutes
99% ("two nines")	3.65 days	21.9 hours	7.31 hours	1.68 hours	14.40 minutes
99.5% ("two nines five")	1.83 days	10.98 hours	3.65 hours	50.40 minutes	7.20 minutes
99.8% ("two nines eight")	17.53 hours	4.38 hours	87.66 minutes	20.16 minutes	2.88 minutes
99.9% ("three nines")	8.77 hours	2.19 hours	43.83 minutes	10.08 minutes	1.44 minutes
99.95% ("three nines five")	4.38 hours	65.7 minutes	21.92 minutes	5.04 minutes	43.20 seconds
99.99% ("four nines")	52.60 minutes	13.15 minutes	4.38 minutes	1.01 minutes	8.64 seconds
99.995% ("four nines five")	26.30 minutes	6.57 minutes	2.19 minutes	30.24 seconds	4.32 seconds
99.999% ("five nines")	5.26 minutes	1.31 minutes	26.30 seconds	6.05 seconds	864.00 milliseconds
99.9999% ("six nines")	31.56 seconds	7.89 seconds	2.63 seconds	604.80 milliseconds	86.40 milliseconds
99.99999% ("seven nines")	3.16 seconds	0.79 seconds	262.98 milliseconds	60.48 milliseconds	8.64 milliseconds
99.999999% ("eight nines")	315.58 milliseconds	78.89 milliseconds	26.30 milliseconds	6.05 milliseconds	864.00 microseconds
99.9999999% ("nine nines")	31.56 milliseconds	7.89 milliseconds	2.63 milliseconds	604.80 microseconds	86.40 microseconds

Availability is one of the most fundamental requirements for system design. You've probably heard the term "high availability" a lot in tech talks. People like to boast about their service being "highly available". But what does it mean exactly?

What is Availability?

High availability refers to the ability of a system to operate continuously, without failure, for a significantly long period. It's a crucial element of system design that ensures systems can provide services to users as intended without unplanned downtimes.

Let's simplify with a pizza example. If it's open when you expect it to be and serves you pizza whenever you want, then it has high availability. In computer systems, this means the system is working and can be used whenever it's needed.

Measuring Availability

Availability is often defined as the percentage of time that a system is operational and able to provide service as expected.

If your favorite pizza place is open 24/7 then it's 100% availability. 100% availability is virtually impossible in a computing environment.

Here's the formula to calculate availability:

$$\text{Availability (\%)} = (\text{Total operational time} / \text{Total time}) \times 100$$

Here are the steps on how to calculate it:

1. Identify the Total Time: This is typically the total amount of time for which you're evaluating the system's availability. It could be a day, week, month, year, etc.
2. Identify the Downtime: This is the total amount of time during the period under evaluation that the system was not operational. This could be due to system crashes, maintenance, network issues, etc.
3. Calculate the Operational Time: This is done by subtracting the downtime from the total time.

$$\text{Total Operational Time} = \text{Total Time} - \text{Downtime}$$

4. Calculate Availability: Divide the operational time by the total time, then multiply the result by 100 to get a percentage.

$$\text{Availability (\%)} = (\text{Total Operational Time} / \text{Total Time}) \times 100$$

For example, if you are evaluating system availability over the course of a year (which has 3652460 = 525,600 minutes), and the system was down for a total of 500 minutes during that time:

$$\text{Total Operational Time} = 525,600 \text{ minutes} - 500 \text{ minutes} = 525,100 \text{ minutes}$$

$$\text{Availability (\%)} = (525,100 \text{ minutes} / 525,600 \text{ minutes}) \times 100 \approx 99.905\%$$

This means the system was available approximately 99.905% of the time.

What is “High” Availability?

Now, how good is 99.905% availability? Is it considered “highly” available?

High availability is often talked about in terms of "nines". If something has "five nines" (99.999%) availability, it's like saying your pizza place is open 99.999% of the time. This means it's only closed about 5 minutes a year! That is a pretty highly available system. Here's a table from [Wikipedia](#) showing availability percentage to the downtime to give you a feel of how the number of 9s would require.

Availability %	Downtime per year ^[note 1]	Downtime per quarter	Downtime per month	Downtime per week	Downtime per day (24 hours)
90% ("one nine")	36.53 days	9.13 days	73.05 hours	16.80 hours	2.40 hours
95% ("one nine five")	18.26 days	4.56 days	36.53 hours	8.40 hours	1.20 hours
97% ("one nine seven")	10.96 days	2.74 days	21.92 hours	5.04 hours	43.20 minutes
98% ("one nine eight")	7.31 days	43.86 hours	14.61 hours	3.36 hours	28.80 minutes
99% ("two nines")	3.65 days	21.9 hours	7.31 hours	1.68 hours	14.40 minutes
99.5% ("two nines five")	1.83 days	10.98 hours	3.65 hours	50.40 minutes	7.20 minutes
99.8% ("two nines eight")	17.53 hours	4.38 hours	87.66 minutes	20.16 minutes	2.88 minutes
99.9% ("three nines")	8.77 hours	2.19 hours	43.83 minutes	10.08 minutes	1.44 minutes
99.95% ("three nines five")	4.38 hours	65.7 minutes	21.92 minutes	5.04 minutes	43.20 seconds
99.99% ("four nines")	52.60 minutes	13.15 minutes	4.38 minutes	1.01 minutes	8.64 seconds
99.995% ("four nines five")	26.30 minutes	6.57 minutes	2.19 minutes	30.24 seconds	4.32 seconds
99.999% ("five nines")	5.26 minutes	1.31 minutes	26.30 seconds	6.05 seconds	864.00 milliseconds
99.9999% ("six nines")	31.56 seconds	7.89 seconds	2.63 seconds	604.80 milliseconds	86.40 milliseconds
99.99999% ("seven nines")	3.16 seconds	0.79 seconds	262.98 milliseconds	60.48 milliseconds	8.64 milliseconds
99.999999% ("eight nines")	315.58 milliseconds	78.89 milliseconds	26.30 milliseconds	6.05 milliseconds	864.00 microseconds
99.9999999% ("nine nines")	31.56 milliseconds	7.89 milliseconds	2.63 milliseconds	604.80 microseconds	86.40 microseconds



But, how many nines are good enough for my system, or is it even necessary? To answer that we first have to discuss Service Level Agreement (SLA).

Service Level Agreement (SLA)

A Service Level Agreement (SLA) is a formal, written agreement between a service provider and a customer that sets expectations regarding the level of service the provider will deliver. This often includes specific metrics and targets the service provider agrees to meet, such as system uptime, response time, and resolution time.

For example, an SLA might stipulate that a system will have an uptime of 99.9% (three nines), and any breach of this may result in penalties, often in the form of service credits to the customer.

SLAs are important because they clearly establish expectations for both the service provider and the customer, offer a standard to measure service performance, and set out remedies or penalties if the service levels aren't met.

For example, here's a screenshot from [Google Cloud's compute engine service's SLA](#):

Compute Engine Service Level Agreement (SLA)

During the Term of the agreement under which Google has agreed to provide Google Cloud Platform to Customer (as applicable, the "Agreement"), the Covered Service will provide a Monthly Uptime Percentage to Customer as follows (the "Service Level Objective" or "SLO"):

Covered Service	Monthly Uptime Percentage
Instances in Multiple Zones	>= 99.99%
A Single Instance	>= 99.5%
Load balancing	>= 99.99%

If Google does not meet the SLO, and if Customer meets its obligations under this SLA, Customer will be eligible to receive the Financial Credits described below. Monthly Uptime Percentage and Financial Credit are determined on a calendar month basis per Project or, for a Single Instance, per instance. This SLA states Customer's sole and exclusive remedy for any failure by Google to meet the SLO. Capitalized terms used in this SLA, but not defined in this SLA, have the meaning set forth in the Agreement. If the Agreement authorizes the resale or supply of Google Cloud Platform under a Google Cloud partner or reseller program, then all references to Customer in this SLA mean Partner or Reseller (as applicable), and any Financial Credit(s) will only apply for impacted Partner or



It specifies the availability uptime in different scenarios and what Google offers to compensate the customers if it fails to achieve them.

Cloud providers typically publish SLAs for their services:

- [Google Cloud SLA](#)
- [AWS SLAs](#)
- [Azure services' SLAs](#)

Note that availability is not the only property that can be specified in an SLA. For example, durability might also be specified as part of the provider's commitments. For example, Amazon S3, the cloud storage service from Amazon Web Services (AWS), boasts an impressive [data durability of 99.999999999% \(eleven nines\)](#). This essentially means that if you store 10,000,000 objects in Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years. We will discuss more about durability in the coming articles.